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Understanding How Exposure to Science Content on Social Media Shapes Normative Expectations of Scientists' Communication: A Social Media Affordances Perspective

Abstract

As social media continue to shape where science is communicated and how it is consumed, this study adopts an affordances approach to advance our understanding of how audiences observe and make sense of online scientific content and evaluate science and scientists more broadly. By analyzing a two-wave survey ($N = 878$) conducted in February and March 2025, this study reveals that online science content exposure is associated with increased normative expectations across a range of scientists' communication behaviors, including political advocacy, warmth- and competence-oriented strategies, and public engagement. Additionally, stronger expectations of competency-oriented communication and political advocacy were associated with greater trust in and vulnerability towards scientists. Political ideology was found to moderate the relationship between science observation and certain normative expectations. Findings underscore the role of social media in influencing science communication norms and (re)shaping what it means to be a trustworthy scientist in an increasingly visible and networked media environment.

Keywords: *social media, affordances, science communication, trust, norms, scientists*

Introduction

In the last few decades, social media platforms have rapidly transformed how people access and engage with information about science, environmental, and health-related issues (Rainie et al., 2015). These platforms operate as dynamic spaces where audiences engage with scientific information, negotiate its meaning within broader cultural contexts, and build and shape their relationships with science and scientists (e.g., Zhang & Lu, 2025). So, too, social media have become powerful tools for scientists and other communicators to share their research, interact with diverse audiences, and foster (scientific) deliberation and dialogue (Yeo, 2015).

A significant body of research has already focused on how scientists can strategically leverage social media for science communication and to promote trust in science (Hwong et al., 2017). Less work, however, has examined the reciprocal side of that engagement; that is, how might the use of these same platforms correlate with *audience's* normative expectations of scientists' communication, roles, and behaviors in these spaces? After all, social media can offer audiences unprecedented access to the informal, behind-the-scenes, and affective dimensions of science, making visible behaviors and knowledge that may have otherwise remained behind lab doors or hidden within the "ivory tower" (Zhang, 2025).

Through social media affordances such as visibility, platforms constitute a particular sociotechnical environment in which scientists' communication styles, disclosures, and interactions are rendered publicly (or sometimes even highly) visible and enduring (Treem et al., 2020). Over time, these repeated exposures can enable and constrain what audiences come to see as normative, appropriate, or expected behaviors for scientists in public life. In this way, social media can contribute to "the set of expectations, schemas, and scripts that [audiences] draw on" when evaluating scientific content and engaging with scientists (Bayer et al., 2020, p. 475).

Therefore, the present study adopts a social media affordances perspective to investigate the potential for social media to (re)shape these normative expectations, as well as the potential consequences of these perception. Such an approach is particularly relevant in today's information landscape, where audiences on platforms like TikTok, Instagram and X are increasingly exposed to different, often informal, models of science communication (e.g., Zeng et al., 2020).

By utilizing panel survey data conducted in February and March 2025, this study seeks to examine whether science observation on social media may be correlated with normative expectations across four dimensions: political expression, warmth-oriented strategies, competence-oriented strategies, and public engagement. Subsequently, these normative expectations may be associated with broader attitudes toward science, particularly so as attitudes towards science become more politically polarized (Gauchat, 2012). Thus, we also examine whether these norms are associated with trust in scientists and willingness to be vulnerable to scientists, and whether political ideology may play a role in moderating these relationships.

Science Observation on Social Media: An Affordances Approach

The concept of affordances was first coined by Gibson (1977) to refer to the potential perceived actions that an environment offers individuals. Later, the concept was applied to communication technology to describe the “relational structure between an object/technology and the user that enables or constrains potential behavioral outcomes in a particular context” (Evans et al., 2017, p. 36). The affordances approach provides a holistic and nuanced perspective towards examining technology and user behavior that transcends specific platforms or features (Evans et al., 2017; Treem & Leonardi, 2013). Rather than assume platform structures exert uniform or predetermined outcomes, such an approach instead highlights the mutuality between a user's subjective interpretations and intentions and a platform's features and technological capabilities to

consider the “possibilities for action” (Evans et al., 2017, p. 36; Majchrzak et al., 2013; Treem & Leonardi, 2013).

Already, a breadth of literature has explored how social media affordances can facilitate interaction and promote participation, engagement, and learning in the realm of political and organizational communication (Kim & Ellison, 2022; Treem & Leonardi, 2013). For example, Kim and Ellison (2022) found that affordances like visibility, persistence, association, and editability can allow people to learn about political engagement by observing others' behaviors on social media. Drawing on Bandura's (1977) social learning theory, the authors argue that social media users can acquire these new norms and behaviors through attention, retention, and reproduction of observed (political) actions of their networks (Bandura, 1977; Kim & Ellison, 2022).

In the context of the present study, then, affordances can perhaps provide a useful high-level lens to examine how social media can similarly structure the possibilities for how audiences observe science, and how that observation can shape normative expectations about science communication. While some emerging research has applied the affordances lens to understand how scientists engage with the public on social media (for example, during the Covid-19 pandemic; Hara & Chae, 2025), this work seeks to extend this perspective to examine how affordances may also shape how *audiences* come to observe, interpret, and form expectations based on that engagement. After all, just as observing other users' political behaviors can influence what users internalize as acceptable or expected political participation norms (Kim & Ellison, 2022), observation of science content on social media may also be associated with beliefs and expectations about what constitutes appropriate or normative science communication.

Here, science content is defined broadly to reflect the diversity of science information and discourse that can be observed on social media, ranging from reports of scientific findings, to personal commentary on or political expression towards science-related topics, to even science misinformation. Such content may originate from a wide range of sources, including scientists, institutions, influencers, or bad-faith actors, and may vary in tone, accuracy, intent, and framing. And while the present study does not directly measure perceived affordances, it conceptualizes science observation as a function of exposure to this broad landscape of science content on social media, the nature of which may be shaped by social media affordances.

An exhaustive list of social media affordances here would be, as Theochraris et al. (2022) argue, “long, subject to some debate, and vulnerable to growing outdated rapidly” (p. 793). Instead, this study identifies four key, top-level affordances shared by many popular social media platforms—visibility, social presence, association, and personalization—that are relevant to science communication and theorize how they may impact how social media might shape science observation, and why that matters for shaping normative expectations.

On social media, users can be afforded the ability to make publicly visible their behaviors, knowledge, preferences and network connections that otherwise would remain inaccessible or invisible (Treem & Leonardi, 2013; Treem et al., 2020; Ellison & Vitak, 2015). For example, people can now observe or easily locate information about other users' attitudes towards science or online science engagement, which can be made visible through their profiles, status updates, and photos (e.g., Halpern et al., 2017). In this way, the affordance of *visibility* can shape what kinds of science content are made visible on social media, that is, not only formal scientific press releases or findings, but also personal commentary on scientific issues, science-related political expression, or more from users' own social media networks.

At the same time, many scientists use social media to present themselves, discuss their research, engage with other users, and respond to current events (Jarreau et al., 2019; Zhang & Lu, 2025). Prior to social media, lay audiences were not privy to such observations, which were often filtered through traditional media or institutional channels (Fahy & Nisbet, 2011). Visibility, then, can quite literally render visible dimensions of scientific work and identity that traditionally remained “in the ivory tower,” thereby providing users with the opportunity to observe and learn how scientists engage and behave on social media.

Furthermore, this visibility is likely amplified by the scale and nature of scientists' engagement: hundreds of thousands of scientists actively maintain social media presences, often with large and heterogeneous audiences (Costas et al., 2020). Many of these scientists also use a single account for both professional and personal expression, collapsing the boundaries between the persona of the scientist and the social media user (Bowman, 2015). Subsequently, audiences may encounter and observe the affective, informal, and interpersonal dimensions of science communication on social media (Zhang, 2025). In doing so, visibility not only influences public exposure to science and scientists, but also structures what kinds of recurring science content and communicative behaviors are observed and potentially internalized as normative.

Relatedly, social media can also promote social presence, which refers to users' subjective feelings of an “awareness of the other person's presence in online environments” (Short et al., 1976, p. 65; Fox & McEwan, 2017). As science content—and the behaviors and knowledge of the people behind it—become increasingly visible on social media, audiences can observe various self-disclosures, storytelling, or interpersonal messaging in science content. Observation of such content, as well as the use of interactive platform features (e.g., tagging, commenting, direct messaging with scientists), can contribute to a heightened sense of social presence (Zhang & Lu,

2024). That is, audiences are afforded an opportunity to feel such mediated closeness to science communicators (e.g., scientists) and perceive them as more “real” and less “removed” from the everyday experiences of non-scientists.

As an affordance, social presence is particularly significant given the persistent gap between lay audiences and the scientific community: a 2021 national poll found that 72% of Americans could not name a single living scientist, a figure that, while still high, represents a nearly 10% decrease from the same survey conducted in 2017 (Research!America, 2017; 2021). While social media has, of course, existed well before both polls, the Covid-19 pandemic appears to have marked a turning point: scientists, science influencers and communicators, and science content became more prominent in public discourse (Fahy & Lewenstein, 2022; Saks & Tyson, 2022). Amid these shifts, social media can contribute to this psychological phenomenon in which users can feel as if they are with other salient social actors (in this case, the scientists) within this technologically-mediated environment (Weidlich & Bastiaens, 2019).

Next, the affordance of *association* allows for the capacity to create and navigate connections between users, content, and networks (Treem & Leonardi, 2013; Fox & McEwan, 2017). Platforms like X, Instagram, and TikTok allow users to connect with other users regardless of whether that other user follows them back; in doing so, users can observe (science) content directly or encounter such posts through mutual “friends” and networks (boyd & Ellison, 2007; Fox & Moreland, 2015). Furthermore, as Ellison et al. (2015) articulate in the context of knowledge-sharing practices within organizations, the association affordance enables users to more easily identify or contact other users with certain expertise (Ellison et al., 2015). These linkages therefore can allow users to situate science content and scientists within broader social and informational networks, help them identify (scientific) expertise, or even shape how audiences

interpret scientists' credibility and relevance. At the same time, users' own networks can shape the type of content, as well as communicative behaviors, they are most likely to observe (Ellison & Vitak, 2015), thus influencing how users might learn and form expectations about science communication.

Lastly, *personalization* can be conceptualized as both receiver-oriented (that is, users customize what they see based on who they interact with) and sender-oriented (how communicators craft content directed towards particular audiences) (Fox & McEwan, 2017; Malhotra, 2024). Social media platforms can afford both: science communicators can personalize their communication style, tone, or content to align with the perceived preferences or expectations of a given audience segment (Luzón, 2013). Users who perceive that content as directed towards “people like them” may be more inclined to interact with such tailoring (Lou et al., 2021). In turn, this type of content may be more likely to appear in their feed and reinforce its perceived typicality. Such reciprocal forms of personalization therefore influence the kinds of science content that are most observable (and memorable) to audiences. Put simply, content that aligns with user preferences may be more likely to be observed and interpreted as what science communication *should* look like on social media.

Normative Expectations of Scientists' Communication on Social Media

Altogether, such an approach can help explain why and how social media may uniquely structure science observation, and offer an framework towards how normative expectations of scientists' communication on these platforms can be shaped. Normative expectations refer to individuals' subjective expectations of norms, which are the behaviors and attitudes that are considered acceptable, appropriate, or expected within a given context (Ajzen, 1991). These expectations relate to the standards or assumptions individuals hold regarding how others should

behave (Burgoon, 2015). Importantly, though, these normative expectations may not be necessarily grounded in reality; rather, they can be shaped by users' "unique and local experiences" (Tankard & Paluck, 2016, p. 182), such as their social media networks or what is made visible (and thus observable) to them through public behaviors and signals (Ross et al., 1977; Fiske & Taylor, 1984).

The affordances lens can help explicate the sociotechnical aspects of the broader communication environment under which these normative expectations can be shaped. As audiences observe science content and recurring patterns of behaviors or communicative styles, these practices may become perceived as normative and therefore a reality, signaling the "normative" or "expected" way for scientists to behave on these platforms (Bayer et al., 2020; Zillich & Wunderlich, 2024). We focus specifically on normative expectations of scientists' communication for two reasons. First, although not all science content on social media originates from scientists, patterns observed across the broader information landscape may still inform or contribute to what audiences come to expect of scientists' roles and behaviors in public discourse (e.g., Pierce (2009) describes a spillover effect in the form of inadvertent emergence of new (shadow) norms in the context of privacy). Second, perceived violations of communication norms may undermine trust or amplify skepticism (Bormann et al., 2022). As scientists seek to position themselves as prominent science communicators with expertise and authority in these online information environments (Huber et al., 2019), identifying these norms and how they might be shaped is thus essential, particularly so in a political climate that has become increasingly hostile towards scientists (Tollefson et al., 2025).

This study focuses on four types that reflect salient aspects of contemporary science communication: scientists' political expression, warmth-oriented strategies, competency-oriented

strategies, and public engagement behaviors. First, we select *political expression* (defined as “practice of posting about political processes or public affairs on social media sites, to which a set of users are given access,” Kim et al., 2021, p. 1124) to capture growing tensions and evolving norms about scientists' political expression on social media, especially in the face of increasing politicization and polarization of scientific issues (Gauchat, 2012). The next two expectations draw from the Stereotype Content Model (SCM), which posits warmth (e.g., friendliness, relatability) and competence (e.g., intelligence, capability) as the two dimensions that guide social perception (that is, how people evaluate others; Fiske et al., 2002). Warmth-oriented strategies include a range of verbal, visual, and aural cues that convey friendliness, relatability, and authenticity; competency-oriented strategies can utilize similar cues to express expertise and intelligence (Suk et al., 2026; Cuddy et al., 2008). Finally, we include *public engagement* to reflect broader expectations for engagement and interaction, particularly as models of science communication move beyond one-directional, “deficit model” approaches (Sturgis & Allum, 2004). Thus, public engagement behaviors refer to those that encourage engagement and interactions, such as replying to comments, answering questions from lay audiences, promoting audience participation, or responding to feedback.

Taken together, we predict that science observation on social media is associated with normative expectations surrounding these four dimensions of science communication due to the affordances of social media. Therefore, we hypothesize the following:

H1. Science observation on social media (Wave 1) will be positively associated with normative expectations of scientists' (a) political expression, (b) use of warmth-oriented strategies, (c) use of competency-oriented strategies, and (d) public engagement on social media (Wave 2).

Implications of Normative Expectations

Normative expectations have long been shown to be powerful “vehicles for social change” that can shape attitudes and behaviors, across fields from social psychology to communication to public policy (Tankard & Paluck, 2016, p. 102). After all, people are motivated to understand the norms of the communities they engage with or belong to, so as to feel a sense of belonging, avoid social rejection, and to maintain accuracy about (social) facts (Cialdini & Goldstein, 2004). People’s normative perceptions, even if objectively inaccurate, can thus become a reality that shapes and guides their own behaviors as well as expectations for others’ behaviors (Ross et al., 1977). Subsequently, many strategic communication campaigns have sought to influence normative expectations instead of attitudes themselves, since these expectations may be more malleable than well-formed attitudes tied to personal ideologies or values (Tankard & Paluck, 2016).

Given the influence that normative expectations can have on how individuals interpret behaviors and evaluate others, we expect normative expectations of scientists’ communication on social media can similarly be correlated with important outcomes: specifically, trust in scientists and willingness to be vulnerable to scientists. Trust in scientists, often conceptualized as an integrative perception of competence, integrity, benevolence, and openness, has important implications for a healthy and informed society (Besley et al., 2020; Cologna et al., 2025). Higher trust provides a foundation for effective and efficient evidence-based decision-making (Algan et al., 2021) and can encourage individual and collective action on pressing scientific challenges like climate change (Cologna & Siegrist, 2020). Yet, trust in scientists (that is, expectations of trustworthiness) alone may not be sufficient, particularly in politicized or polarized scientific contexts where individuals must not only see scientists as trustworthy, but also be willing to cede

authority or accept potential risks based on scientific recommendations (Brossard & Nisbet, 2007; Cologna et al., 2025). The second outcome then focuses on this more behavioral form of trust; that is, the willingness to make oneself vulnerable to scientists, where vulnerability might be understood as “a willingness to take personal advice from scientists or want government to take advice from scientists” (Besley et al., 2020, p. 710).

Although few studies have directly examined how normative expectations influence these outcomes, there is growing evidence that the underlying behaviors themselves—political expression, warmth, competence, and public engagement—can influence trust in and willingness to be vulnerable to scientists. For example, a majority of German and U.S. citizens support and expect scientists' political advocacy on climate change (Cologna et al., 2021); this advocacy is associated with increased public trust, especially for liberal individuals (Cologna et al., 2024). Similarly, when scientists engage in strategies that influence the extent to which they are perceived as competent and warm, people may be more likely to support science policies (Zhang & Lu, 2025). Lastly, public engagement has long been explored as a means for restoring trust in science (Wynn, 2006).

Still, equally important is whether audiences perceive those behaviors as normative, because these normative expectations can operate as heuristics of sorts for interpreting behaviors, particularly so in contexts where individuals lack motivation and ability or are faced with a novel or unfamiliar situation (Fiske & Taylor, 1984; Petty et al., 2002). Scientists on social media (still a relative novelty in and of itself), then, is one such context: when forming judgements related to trust and vulnerability in watching a science TikTok video, for example, audiences may choose to evaluate whether that scientist's communication styles or behaviors align with their own normative expectations instead of assessing them based on their message or expertise alone (Cialdini & Trost,

1998). In other words, these normative expectations become cues for assessing scientists' typicality, and by extension, their trustworthiness. However, given limited empirical investigation, we ask:

RQ1: How might normative expectations of scientists' communication on social media be associated with (a) trust in scientists and (b) willingness to be vulnerable to scientists?

The Role of Political Ideology as a Moderator

In recent decades, ideologically-driven differences in trust in scientists and other attitudes towards science have widened in the U.S., with conservatives expressing lower trust in scientists and increased skepticism towards scientists and scientific institutions than liberals (Gauchat, 2012; Kennedy & Tyson, 2023). These divisions, then, may extend to how people respond to scientists' behaviors and communicative strategies on social media.

Indeed, people are more likely to attend to and believe information that aligns with their prior beliefs, while discounting or resisting incongruent information (Taber & Lodge, 2006). For conservatives, behaviors like political expression or informal, personalized messaging may be viewed as dissonant messages, and thus could be less likely to shift their normative expectations and instead reinforce negative attitudes (Nisbet et al., 2015). On the other hand, liberals, who tend to hold more positive attitudes toward scientists and may be more supportive of scientists' public engagement and advocacy roles, might be more receptive to these same behaviors and perceive them as normative (Funk et al., 2019).

Furthermore, selective exposure and political homogeneity on social media may mean that conservatives and liberals differ in the extent to which they encounter science content on social media (e.g., Garrett, 2009). However, these patterns may be complicated by algorithmic curation (that is, algorithms might promote the spread of content outside of one's ideological preferences;

Thorson & Wells, 2016) or by the socialization of social media information (e.g., via social endorsements), which can promote exposure to information from politically-heterogenous sources (Messing & Westwood, 2014). Thus, even when individuals do observe the same set of interactions or content (i.e., observing scientists reply to comments or advocate for policy), the extent to which their observations might be related to their normative expectations may be moderated by their political ideology.

However, a number of studies have suggested that political ideology may not actually moderate these relationships, or perhaps to a lesser extent: a 2024 study examining the associations between social media use and perceptions of social norms related to the Covid-19 vaccine found no significant interaction with political affiliation (Chinn & Hasell, 2024). Others, too, have found a lack of moderation effects even in politically-polarizing contexts such as climate change (e.g., Hart et al., 2023).

Still, investigating political ideology as a moderator allows us to examine whether normative expectations about scientists' communication behaviors are malleable even for partisan audiences, who might hold otherwise stable and affect-laden attitudes toward science and scientists (Gauchat, 2012). As described earlier, social media allows for new opportunities for intentional and incidental science observation; if such observations can indeed shift these normative expectations, even among partisans, these findings can further clarify how everyday, online encounters with science might influence ideological gaps in trust in science or encourage (or discourage) broader receptivity towards scientific recommendations. Given these conflicting findings, we ask as the last research question:

RQ2: Does political ideology moderate the association between science observation on social media and normative expectations of scientists' communication on social media?

Method

Data

The present study analyzes original data from a two-wave panel survey of U.S. participants collected in February and March of 2025. Participants were recruited through Forthright, an online sampling platform, using quota sampling to closely reflect U.S. census characteristics. Only participants who reported using social media and currently possessed an active account were eligible to participate, since the independent variable required participants to use social media.

Wave 1 data were collected between February 19 and February 27, 2025, yielding 1,268 valid responses after excluding participants who failed either of two attention check questions embedded within the survey flow. Of those initial respondents, 878 completed the second wave of the survey (conducted between March 17–21, 2025), resulting in a retention rate of 69.2%. Wave 2 respondents were broadly representative of the sample in Wave 1, with no demographic variable deviating by more than two percentage points across waves (see **Appendix A**).

The average age of participants was 47.6 years ($SD = 16.6$). Approximately half (49.0%) identified as female, 47.4% as male, and 3.6% as nonbinary or chose not to disclose their gender. The majority of participants were White (67.4%), followed by African American (12.1%), Hispanic or Latine (11.5%), Asian (4.2%), and 5.0% identified as “other.” For education, 10.5% of participants held an advanced degree, 27.1% held a Bachelor’s degree, 40.1% had completed some college or held an associate degree, 20.3% were high school graduates, and 2.0% had not completed high school. The median self-reported family income was \$50,000 - \$74,999.

Measures

The survey procedure and items for each measure are included in **Appendix A**.

Science Observation on Social Media

In the first wave of the survey, participants were asked, “Generally, how often have you come across the following types of information about science on social media?” The 8 types of science content included items such as “reports on new discoveries, research findings, or expert analyses,” and “posts encouraging political action on science-related issues.” ($M = 2.59$, $SD = 0.81$, $\alpha = 0.88$).

Normative Expectations of Scientists' Communication on Social Media

In the second wave, participants were asked to indicate their agreement (1 = *strongly disagree*, 5 = *strongly agree*) with several statements related to their expectations for scientists' communication on social media across four types of expectations. These statements are drawn from existing research (e.g., Suk et al., 2026; Zhang, 2025) that have identified common strategies utilized by scientists on social media through content analyses. Given the lack of established constructs and scales related to these expectations, we wanted to explore the extent to which items in each type of expectations indeed loaded onto a single, distinct factor. An exploratory factor analysis (EFA) using Principal Component Analysis with Varimax rotation indicated good suitability of the data for factor analysis ($KMO = 0.89$; Bartlett's $\chi^2 (136) = 4346.07$, $p < 0.001$) and identified four factors that explained 56.4% of the total variance (public engagement: 18.8%, competence strategies: 13.7%, warmth strategies: 12.7%, and political expression: 11.2%). **Appendix B** provides the full EFA results.

(1) *Public engagement* was captured by asking participants to indicate their agreement with four statements, such as “replying to comments and questions from the public about science” and “responding to feedback and encouraging audience participation” ($M = 3.97$, $SD = 0.94$, $\alpha = 0.83$).

(2) *Competence-Oriented Strategies* were measured with six items, such as “using precise, technical language” and “prioritizing accuracy and evidence-based communication” ($M = 4.02$, $SD = 0.47$, $\alpha = 0.69$).

(3) *Warmth-Oriented Strategies* were similarly measured with five items, including “incorporating humor, memes, or pop culture references” and “using a conversational, friendly tone” ($M = 3.67$, $SD = 0.61$, $\alpha = 0.63$).

(4) *Political expression* was captured by asking participants to indicate their agreement with two statements, “express their opinions on political and societal issues unrelated to science” and “express their opinions on political and societal issues related to science” ($M = 3.02$, $SD = 1.01$, $r_{SB} = 0.62$).

Trust in Scientists

In the second wave of the survey, twelve items were averaged to create an index, adapted from Cologna et al., 2025. Examples of these items include, “When I think about scientists, most are...competent / qualified when it comes to conducting high-quality research / honest.” Participants responded on a 5-point scale of 1 = *strongly disagree* to 7 = *strongly agree*. ($M = 3.81$, $SD = 0.72$, $\alpha = .94$).

Willingness to be Vulnerable to Scientists

In the second wave of the survey, three items were averaged to create an index, adapted from Mede et al. (2025). These were, “How much or little should people rely on scientists’ guidance when making lifestyle choices related to science?,” “How much or little should governments rely on scientists’ guidance when making decisions related to science?,” and “How much or little control do you want scientists to have on government decisions related to science?”

Participants responded on a 5-point scale of 1 = *Very little* to 7 = *Very much*. ($M = 3.70$, $SD = 0.94$, $\alpha = .84$).

Political Ideology

In the first wave of the survey, political ideology was measured by asking participants “When it comes to politics, do you generally consider yourself to be...” on a scale that ranged from (1) Very liberal to (7) Very conservative ($M = 3.69$, $SD = 1.80$).

All items for each scale employed in the study are included in **Appendix B**. Inter-item correlations for all variables are reported in **Table 1**.

[insert Table 1 here]

Results

First, ordinary least squares (OLS) regressions were used to examine how science observation on social media was associated with the four dimensions of normative expectations of scientists' communication on social media (H1a-d), controlling for the following demographic variables: age, gender, race, education, and political ideology in Wave 1 (**Table 2**; see **Appendix A** for tests of regression assumptions). Science interest in Wave 1 was also used as a control variable ($M = 3.41$, $SD = 1.01$), as it may relate to the variables of interest in the current study. Results showed a positive and significant main effect of science observation on social media on normative expectations of scientists' political expression ($B = 0.15$ (0.04), $p < 0.001$), use of warmth-oriented strategies ($B = 0.15$ (0.03), $p < 0.001$) and competence-oriented strategies ($B = 0.06$ (0.03), $p = 0.012$), as well as public engagement behaviors ($B = 0.13$ (0.04), $p < 0.001$). Thus, H1a-d were supported.

[insert Table 2 here]

Next, OLS regressions were run to include the interactions of political ideology and science observation on social media to examine how associations with science observation may be moderated by political ideology (RQ2, **Table 3**). Political ideology had a significant interaction with science observation on social media for normative expectations of scientists' use of warmth-oriented strategies ($B = 0.03 (0.01), p = 0.016$). Johnson-Neyman analyses revealed that as science observation on social media increased, so did participants' expectations of scientists' use of warmth strategies; however, these differences were only significant as political ideology increased (>1.80). Put in other words, conservatives who indicated higher science observation had significantly higher expectations for using warmth-oriented strategies than did conservatives who indicated lower science observation. **Figure 1** provides a visualization depicting the conditional effect of political ideology. No other interactions were significant.

[insert Table 3 here]

[insert Figure 1 here]

Additional exploratory analyses were conducted to understand how different types of science content may specifically drive these relationships. To do so, science observation was categorized into three sub-types (*informational*, reflecting content aimed at informing or educating audiences about scientific topics; *advocacy*, encompassing content in which individuals express opinions, promote activism, or encourage political action related to science topics; and *provocative*: consisting of emotionally-charged, polarizing, or misleading science content). Advocacy science observation was associated with stronger normative expectations of political expression and warmth strategies, while provocative science observation was associated with decreased normative expectations of political expression but increased normative expectations of public engagement. The only significant interaction with political ideology emerged for informational science

observation, such that conservatives who indicated increased informational science observation had significantly higher expectations for scientists' use of warmth strategies. Full results and regression tables are reported in **Appendix C**.

A final set of OLS regressions were used to test how normative expectations of scientists' communication on social media were associated with trust in scientists (RQ1a) and willingness to be vulnerable to scientists (RQ1b) in the second wave, again controlling for age, gender, race, education, political ideology, and science interest (**Table 4**). Results showed a positive and significant main effect of both normative expectations of scientists' political expression ($B = 0.17$ (0.02), $p < 0.001$) and use of competence-oriented strategies ($B = 0.32$ (0.05), $p < 0.001$) on trust in scientists. For willingness to be vulnerable to scientists, there was a positive and significant main effect of both normative expectations of scientists' political expression ($B = 0.15$ (0.03), $p < 0.001$) and use of competence-oriented strategies ($B = 0.40$ (0.06), $p < 0.001$).

[insert Table 4 here]

Discussion

In the wake of the COVID-19 pandemic, and amid growing political polarization and hostility towards scientists and scientific institutions in the U.S., scientific content and information has taken on heightened visibility and urgency in public discourse (Saks & Tyson, 2022; Tollefson et al., 2025). In particular, social media platforms have emerged as critical sites for both scientists and lay audiences: a recent study from the Pew Research Center (2022) showed that nearly half (48%) of U.S. adult social media users reported seeing science content on these platforms in the past few weeks. More so, one in three users now say they follow accounts focused on science content (a 7% increase from 2017). So, too, have scientists become increasingly present and visible in these spaces, as they share their research, challenge misinformation, advocate for public policies,

and more (Costas et al., 2020; Zhang & Lu, 2025). As social media platforms continue to shape where science is communicated and how it is consumed, this study offers an examination of how audiences observe and make sense of scientific content in these spaces, and evaluate science and scientists more broadly, by analyzing data from a two-wave U.S. national panel survey conducted in early 2025.

First, we applied a social media affordances approach to investigate how these platforms might contribute to the formation of the very norms that people perceive about how scientists should communicate and interact with the public. Through affordances such as visibility and association, social media can make visible previously-hidden or private information (e.g., scientists engaging in political expression, responding to questions from their networks, or disclosing personal experiences) (Treem & Leonardi, 2012; Kim & Ellison, 2021). Over time, observation of such information and behaviors can lead audiences to perceive them as normative, appropriate, or even expected ways for scientists to communicate (Bayer et al., 2020). Our results offer support for the association between science observation on social media and audiences' normative expectations of scientists' communication behaviors. Specifically, the more frequently audiences observe science content on social media, the more likely they are to perceive scientists' political expression, warmth- and competence-oriented strategies, and public engagement behaviors as typical or normative.

Next, we found that political ideology interacted with science observation on social media, but was only significant for normative expectations of warmth-oriented strategies. In particular, among social media users with low science observation, those who were conservatives and those who were liberal differed significantly in how much they expected scientists to use warmth-oriented strategies (liberals indicated higher expectations than conservatives). But among those

with high science observation, those partisan differences disappeared (controlling for science interest). Put differently, increased science observation among conservatives was associated with a significant increase in normative expectations of warmth-oriented behaviors.

The fact that this pattern only occurred for warmth-oriented strategies holds two interesting practical implications: first, it suggests a unique opportunity for scientists to connect with more skeptical audiences on social media. That is, warmth cues such as humor, personal self-disclosure, or using casual and relatable language may serve as potential inroads for shifting what conservatives expect of, and how they perceive, scientists, in line with growing experimental research (e.g., McKasy et al., 2023). And particularly given that much of science content on social media is encountered incidentally rather than intentionally (Funk et al., 2017), even casual or unintentional observation of warmth-oriented strategies that evoke humanizing portrayals of scientists may help reduce expectations of scientists as distant elites or threats (Chinn et al., 2024). Second, the findings also point to a lower threshold for shifting normative expectations that are interpersonal in nature, as opposed to those related to scientists' political, civic, or expert roles. Still, the broader implication is encouraging: even amid rising anti-intellectualism and polarization of science, normative expectations around scientists' warmth may be amenable to change through repeated, visible exposure, and could possibly serve as a mechanism for bridging ideological divides. Scientists and science communication practitioners may therefore consider incorporating evidence-based affective or interpersonal communication strategies that signal warmth and benevolence (Besley et al., 2021; Zhang & Lu, 2025).

Finally, the analyses found that certain normative expectations are associated with both trust in scientists and willingness to be vulnerable to them. Specifically, normative expectations of competence-oriented strategies and political expression behaviors were correlated with these two

implications. These findings raise a number of considerations and questions. First, why might only two of the normative expectations influence trust and vulnerability? Perhaps such a pattern highlights the distinction between an *expectation* of a behavior as typical or even appropriate and *interpretation* of that behavior as trustworthy. After all, the participatory nature and user-generated content on social media means that users are often exposed to science content that can be both credible *or* non-credible (e.g., performative engagement, misinformation, inflammatory and fear-based appeals, and more) (Chinn et al., 2023). As a result, audiences may view some behaviors as normative but not necessarily rely on them exclusively for trust-related judgements; another possibility is that these behaviors may not have been consistently performed or observed in ways that do signal broader trust in scientists.

Second, that both competence and political expression norms were associated with increased trust and vulnerability towards scientists reflect at once both persistent and emerging normative expectations. On one hand, longstanding stereotypes and expectations of scientists as authoritative experts who emphasize their expertise, use jargon and technical language, and talk primarily about their research may align with normative expectations of competence-oriented strategies (Fiske & Dupree, 2014; Chambers, 1983). On the other hand, political expression represents an evolving yet contested dimension of scientific identity: debates about whether the role of scientists as political advocates undermine credibility and trust continue to this day within the scientific community (Büntgen, 2024; van Eck et al., 2024). And with recent work showing that a majority of both scientists and citizens across the U.S. and Germany support greater political expression and engagement by scientists, especially on urgent issues like climate change (Cologna et al., 2025), future research would benefit from further exploring the relationship between trust in

science and support for scientists to advocate their position politically and in interactions with citizens.

By applying the affordances approach to the science communication context, the present study suggests that social media introduces a different consideration for how science communication norms might emerge and evolve. In contrast to the rather top-down, hierarchical nature of institutional norm-setting, norms on social media can emerge through (relatively more) decentralized and participatory processes. In particular, the findings suggest that audience form their own normative expectations of how scientists should communicate through observation of science content on social media. These normative expectations, then, are no longer exclusively dictated by professional standards but instead can be influenced by observations facilitated through social media affordances such as visibility or social presence. These platform-mediated observations thus help facilitate and enable a different and parallel set of norms, grounded in audience interpretation rather than disciplinary consensus.

This holds implication implications for scientists, science communication practitioners, and social media platforms alike. If the goal is to rebuild public trust and meaningfully engage diverse audiences, then their evolving normative expectations of scientific communication must be acknowledged as central to effective engagement. After all, violations of audience norms can lead to lower levels of trust, diminished support for science, and reduced interest in seeking scientific information (Zhang & Lu, 2025). Similarly, social media platforms and industry decision-makers invested in promoting a healthy and informed public may consider how their algorithms and discovery features can amplify (verified) scientists and science communicators in ways that do not require them to fully conform to social media logics that often prioritize virality and sensationalism over nuance.

Limitations & Future Directions

Of course, this study is not without limitations. First, all measures relied on self-reported survey data, which are susceptible to overstatements of media exposure due to demands on participants' memory (Prior, 2009). Participants' reports of their social media exposure may therefore not accurately reflect the frequency, content, or source of the science-related information they actually encountered. While this does limit claims about how direct exposure to scientists specifically may shape expectations, it also highlights a potential spillover effect: even incidental or general science observation on social media (i.e., from other users) may contribute to emerging norms about how scientists should communicate.

Similarly, future research could directly examine how specific affordances play into the way trust in scientists is arrived at by audiences, such as through qualitative and quantitative approaches that look at how different affordances are associated with science information-seeking and engagement behaviors or platform-sensitive approaches that enable more specificity into how the affordances of platforms are relational and multi-layered (Bucher & Helmond, 2017). So too, additional research would benefit from an explicit comparison of science observation on social media with that on other information sources, such as traditional news media, which would help differentiate the unique impacts of social media on normative expectations.

Finally, while the hypotheses are theoretically derived, this study cannot definitively establish causal direction. The study was part of a larger project on social media and science communication, and while the two-wave panel design sought to establish temporal precedence between science observation and the mediators and outcome variables, the latter were only collected at the second wave. This therefore limited the ability to test lagged or reciprocal effects and to isolate the influence of science observation on normative expectations at Wave 2 by

controlling for those expectations at Wave 1. Thus, future longitudinal work would benefit from measuring both the independent and dependent variables at multiple time points, with longer gaps between waves, to further clarify the nature of the relationships and causal directions between science observation on social media, normative expectations, and trust in scientists. Such work could be further triangulated with experimental approaches or those that leverage large-scale social media data, such as those that examine how users actually engage and interact with posts featuring these behaviors and strategies.

Table 1. Inter-item correlations between variables in the model ($N = 878$).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) Science Observation on Social Media	—						
(2) Public Engagement Behaviors	.207**	—					
(3) Competence-Oriented Strategies	.179**	.548**	—				
(4) Warmth-Oriented Strategies	.236**	.519**	.475**	—			
(5) Political Expression	.250**	.246**	.217**	.397**	—		
(6) Trust in Scientists	.144**	.226**	.356**	.247**	.389**	—	
(7) Willingness to be Vulnerable to Scientists	.211**	.240**	.357**	.249**	.388**	.608**	—

Note. ** Correlation is significant at the 0.01 level (2-tailed).

Table 2. Science observation on social media (wave 1) predicting normative expectations of scientists' communication on social media (wave 2).

<i>Normative Expectations of Scientists' Communication on Social Media</i>								
	Political Expression		Warmth Strategies		Competence Strategies		Public Engagement	
	B	(SE)	B	(SE)	B	(SE)	B	(SE)
Age	-0.006**	0.002	0.002	0.001	0.002 [#]	0.001	0.002	0.002
Gender (Women)	-0.068	0.064	0.069 [#]	0.042	-0.044	0.039	0.058	0.052
Race (White)	-0.125 [#]	0.069	-0.001	0.045	-0.008	0.042	-0.066	0.056
Education	-0.016	0.022	0.011	0.014	0.018	0.013	-0.017	0.018
Political Ideology	-0.164***	0.019	-0.036**	0.012	-0.035**	0.011	-0.039**	0.015
Science Interest	0.151***	0.035	0.050*	0.023	0.098***	0.022	0.112***	0.029
Science Observation	0.150***	0.044	0.151***	0.029	0.063*	0.027	0.133***	0.036
Constant	3.206***	0.209	3.044***	0.136	3.493***	0.127	3.404***	0.169

Note. Unstandardized coefficients reported. [#] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3. Interaction of political ideology and science observation on social media (wave 1) predicting normative expectations of scientists' communication on social media (wave 2).

<i>Normative Expectations of Scientists' Communication on Social Media</i>								
	Political Expression		Warmth Strategies		Competence Strategies		Public Engagement	
	B	(SE)	B	(SE)	B	(SE)	B	(SE)
Age	-0.006**	0.002	0.002	0.001	0.002 [#]	0.001	0.002	0.002
Gender (Women)	-0.068	0.064	0.070 [#]	0.042	-0.044	0.039	0.059	0.052
Race (White)	-0.125 [#]	0.070	0.000	0.045	-0.007	0.042	-0.065	0.056
Education	-0.016	0.022	0.011	0.014	0.018	0.013	-0.018	0.018
Political Ideology	-0.172**	0.061	-0.126**	0.040	-0.076*	0.037	-0.123*	0.049
Science Interest	0.150***	0.036	0.049*	0.023	0.097***	0.022	0.111***	0.029
Science Observation	0.139	0.096	0.018	0.062	0.003	0.058	0.011	0.078
Ideology x Science Observation	0.003	0.022	0.034*	0.014	0.015	0.013	0.031 [#]	0.018
Constant	3.238***	0.316	3.414***	0.205	3.660***	0.192	3.743***	0.255

Note. Unstandardized coefficients reported. [#] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4. Normative expectations of scientists' communication on social media predicting trust in scientists and willingness to be vulnerable to scientists (wave 2).

	Trust in Scientists		Willingness to be Vulnerable to Scientists	
	B	(SE)	B	(SE)
Age	0.003*	0.001	−0.003 [#]	0.002
Gender (Women)	0.036	0.043	−0.033	0.053
Race (White)	0.120**	0.046	0.212***	0.057
Education	−0.002	0.015	0.017	0.018
Political Ideology	−0.128***	0.013	−0.211***	0.016
Science Interest	0.060**	0.022	0.055*	0.027
<i>Normative Expectations of Scientists' Communication on Social Media</i>				
Political Expression	0.167***	0.024	0.150***	0.030
Warmth Strategies	−0.015	0.043	0.004	0.054
Competence Strategies	0.321***	0.045	0.398***	0.056
Public Engagement	−0.019	0.035	−0.007	0.043
Constant	2.189***	0.200	2.194***	0.248

Note. Unstandardized coefficients reported. [#] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

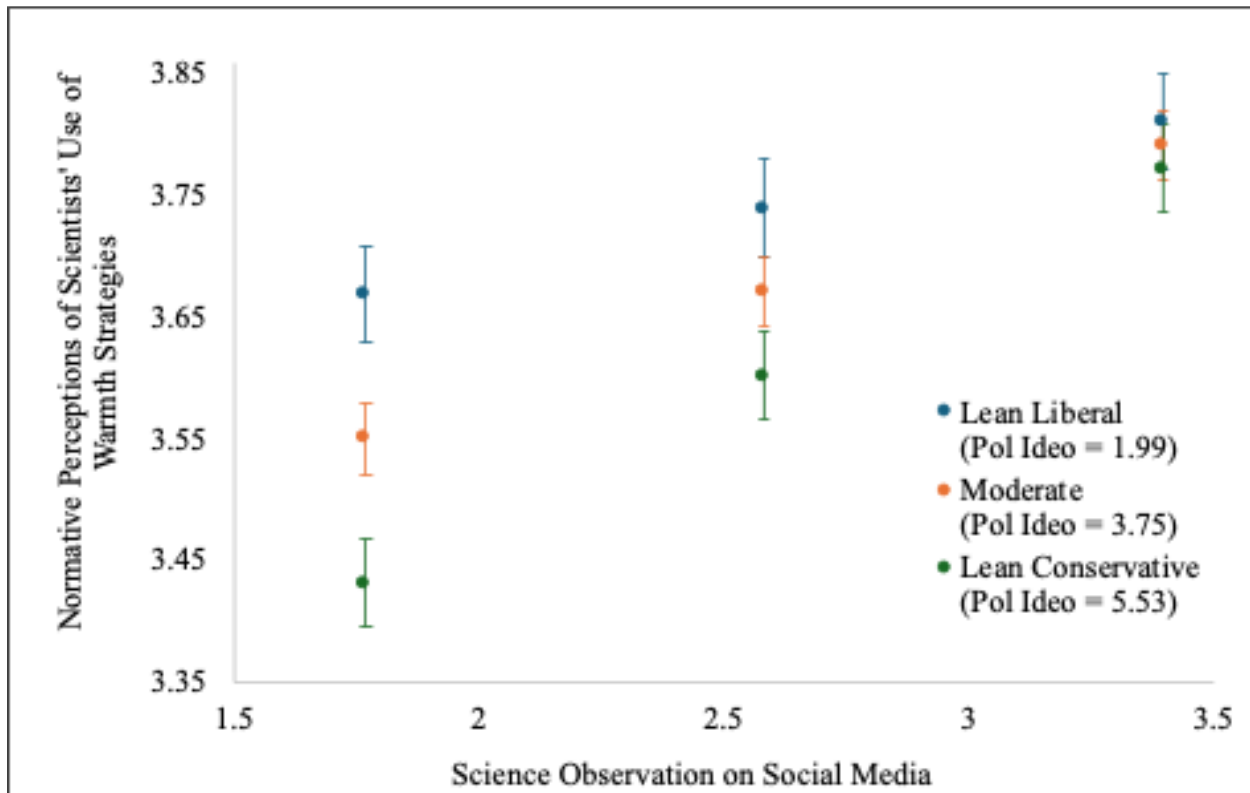


Figure 1. Effect of science observation media on normative expectations towards scientists' use of warmth-oriented strategies. Conditional effects of political ideology are shown at one standard deviation above (lean liberal) and below (lean conservative) the mean (moderate).

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